

CSE113

Foundations of Computer Science

State University of New York, Korea

Plan

- propositional logic and scope of our study
- Truth table
- Logical Equivalence
- Valid Arguments

Propositional Logic and Scope of our study

Proposition

Definition

- A **statement** or **proposition** is a sentence for which a truth value (either true or false) can be assigned

Examples

- The atomic number of Oxygen is 8
- Hangul is made up of 14 consonants and 10 vowels
- There exists life in other planets.
- $((a \rightarrow b) \wedge a) \rightarrow b$
- $(a \wedge \sim a)$

Scope of our study

- Mathematical logic, not ambiguous English
- Compound statements, not unit statements
- So, we will check if a proposition like $(p \rightarrow q) \rightarrow (q \rightarrow p)$ is true or false

Why logic?

Artificial Intelligence 47 (1991) 31–56
Elsevier

31

Logic and artificial intelligence

Nils J. Nilsson

Computer Science Department, Stanford University, Stanford, CA 94305, USA

Received February 1989

Abstract

Nilsson, N.J., Logic and artificial intelligence, *Artificial Intelligence* 47 (1990) 31–56.

The theoretical foundations of the logical approach to artificial intelligence are presented.

- Quote: “Logic provides the vocabulary and many of the techniques needed both for analyzing the processes of representation and reasoning and for synthesizing machines that represent and reason.”

Example: Software intelligence used at FAANG

Question: Simplify this code

5	<code>int x = 0;</code>
6 ▾	<code>while (x < 10){</code>
7	<code> x = x + 1;</code>
8	<code>}</code>

- Answer: x must equals to 10. Following three facts
 - $x < 11$ at Line 6 (before entering the loop)
 - $x \geq 10$ after the loop
 - x is an integer

Truth Tables

Compound statements

Definition

- A **compound statement** is a complex sentence that is obtained by joining **propositional variables** using **logical connectives**

Logical operator	Notation	Read as
Negation	$\sim p$	not p
Conjunction	$p \wedge q$	p and q
Disjunction	$p \vee q$	p or q
Conditional	$p \rightarrow q$	p implies q if p , then q p only if q q if p q , provided that p
Biconditional	$p \leftrightarrow q$	p if and only if q
Logical equivalence	$p \equiv q$	p logically equivalent to q

Examples

- $(p \vee q) \wedge \sim (\sim p \wedge r)$
- $(\sim p \wedge q \wedge r) \vee (q \vee \sim r)$

Truthfulness of compound statements

Negation ($\sim p$)

Definition

- **Negation** of a statement p , denoted by $\sim p$, is a statement obtained by changing the truth value of p .

p	$\sim p$
T	F
F	T

Truthfulness of compound statements

Conjunction ($p \wedge q$)

Definition

- **Conjunction** of statements p and q , denoted by $p \wedge q$, is a statement such that it is true if both p and q are true and it is false, otherwise.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Truthfulness of compound statements

Disjunction ($p \vee q$)

Definition

- **Disjunction** of statements p and q , denoted by $p \vee q$, is a statement such that it is false if both p and q are false and it is true, otherwise.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Truthfulness of compound statements

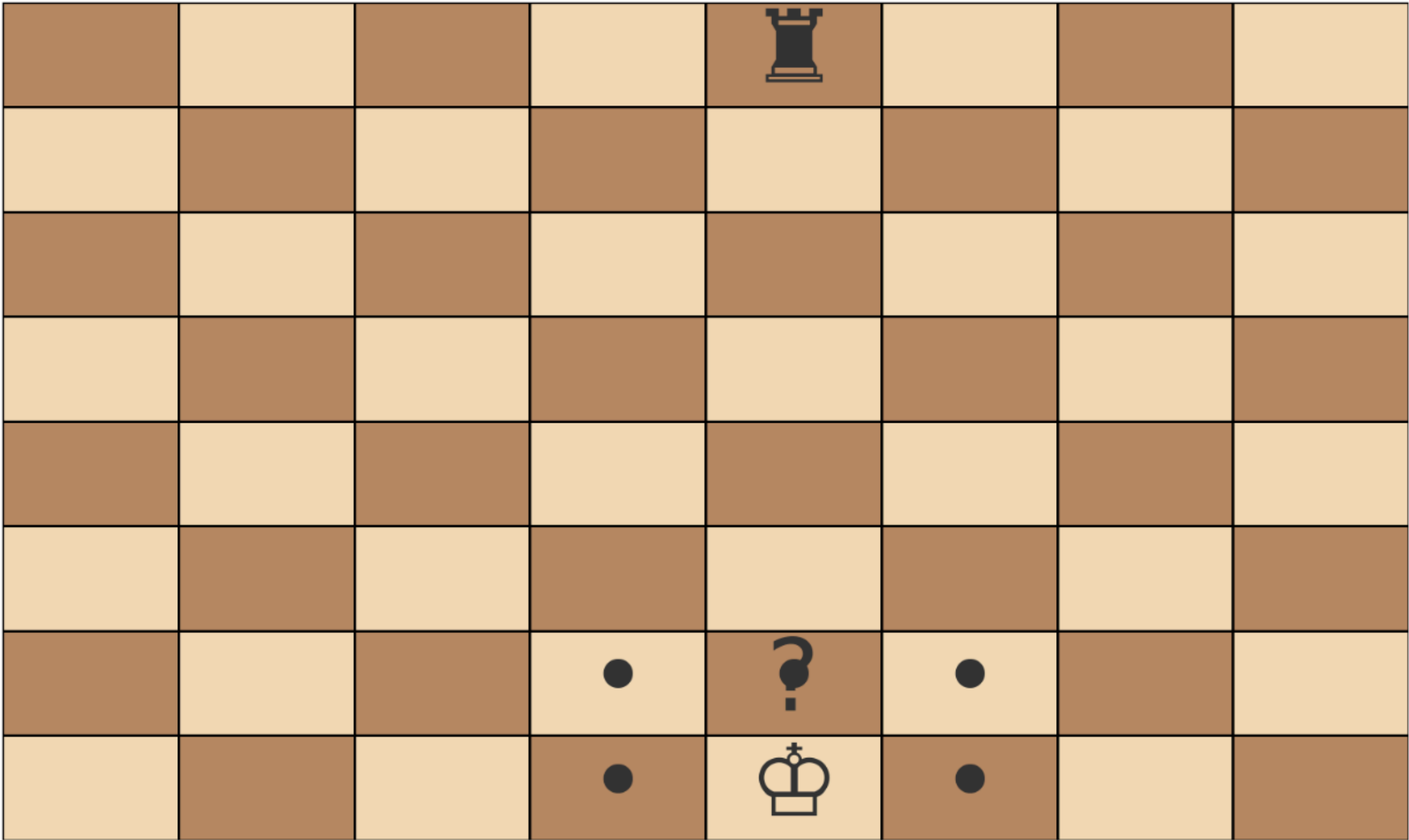
Exclusive or $(p \oplus q)$

Definition

- **Exclusive or** of statements p and q , denoted by $p \oplus q$, is defined as p or q but not both. It is computed as $(p \vee q) \wedge \sim (p \wedge q)$

p	q	$p \vee q$	$p \wedge q$	$\sim (p \wedge q)$	$(p \vee q) \wedge \sim (p \wedge q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Example: "In a chess, when you're in check, you can either move the king out of check or block the check, but not both."



Truthfulness of compound statements

Definition

- **Conditional** or **implication** is a compound statement of the form “if p , then q ”. It is denoted by $p \rightarrow q$ and read as “ p implies q ”. It is false when p is true and q is false, and it is true, otherwise.

$p \rightarrow q$ is defined as
 $\sim p \vee q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Examples on False $\rightarrow$ Anything is true

- If $1+1 = 3$, then $1 = 0$
- If the earth is flat, I am walking on the moon

Truthfulness of compound statements

Biconditional statement ($p \leftrightarrow q$)

Definitions

- The **biconditional** of p and q is of the form “ p if and only if q ” and is denoted by $p \leftrightarrow q$. It is true when p and q have the same truth value, and it is false, otherwise.
- $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Examples

- Assume x and y are real numbers.
“ $x^2 + y^2 = 0$ if and only if $x = 0$ and $y = 0$.”

Precedence of Logical Operators

Priority	Operator	Comments
1	$\sim$	Evaluate $\sim$ first
2	$\wedge$ $\vee$	Evaluate $\wedge$ and $\vee$ next; Use parenthesis to avoid ambiguity
3	$\rightarrow$ $\leftrightarrow$	Evaluate $\rightarrow$ and $\leftrightarrow$ next; Use parenthesis to avoid ambiguity
4	$\equiv$	Evaluate $\equiv$ last

- $p \vee q \wedge r$ reads as ...
- $\sim p \rightarrow q$ reads as ...
- $s \wedge q \rightarrow p$ reads as ...

**Exercise 1: check truthfulness of
 $(p \rightarrow q) \rightarrow (q \rightarrow p)$ with a truth table**

Logical Equivalence

Logic equivalence

Definition

- Two statement forms p and q are **logically equivalent**, denoted by $p \equiv q$, if and only if they have the same truth values for all possible combination of truth values for the propositional variables

Checking logical equivalence

1. **Construct and compare truth tables** (most powerful)
2. Use logical equivalence laws

Logical equivalence: Example

Problem

- Show that $p \wedge (q \vee r) \not\equiv (p \wedge q) \vee r$

p	q	r	$q \vee r$	$p \wedge (q \vee r)$	$p \wedge q$	$(p \wedge q) \vee r$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	F	T
T	F	F	F	F	F	F
F	T	T	T	F	F	T
F	T	F	T	F	F	F
F	F	T	T	F	F	T
F	F	F	F	F	F	F

**Exercise 2: check the logical equivalence
between $(p \rightarrow q)$ and $(\sim q \rightarrow \sim p)$**

Exercise 3

Which of the following statements about logical equivalence is true?

- A. The formulas $p \wedge \text{true}$ and $p \rightarrow \text{true}$ are equivalent.
- B. The formulas $p \wedge \text{true}$ and $p \vee \text{false}$ are equivalent.
- C. The formulas $p \wedge \text{true}$ and $p \oplus \text{true}$ are equivalent.
- D. The formulas $p \vee \text{true}$ and $\text{true} \rightarrow p$ are equivalent.

Midterm Exam I (March 10, 2021, 08:30 am - 09:55 am)
CSE 215: Foundations of Computer Science

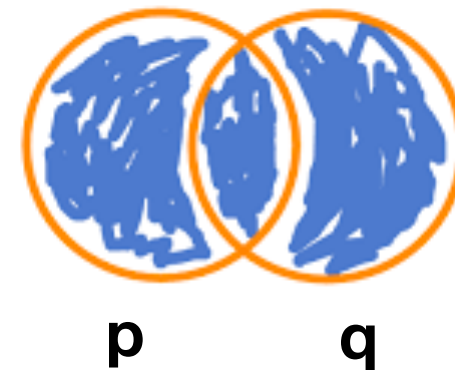
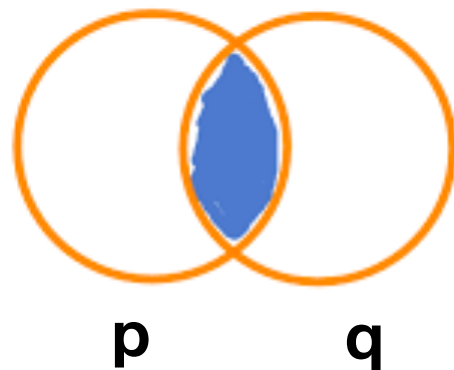
Problem 1. [5 points]

Construct a truth table for the following statement form: $(p \wedge (q \vee r)) \rightarrow (p \wedge r)$.

Equivalence Laws

Commutative Law

Laws	Formula	Formula
Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$



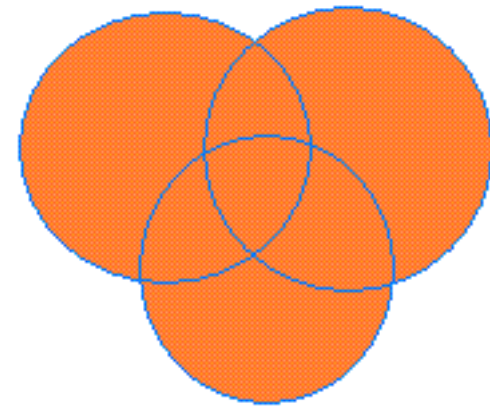
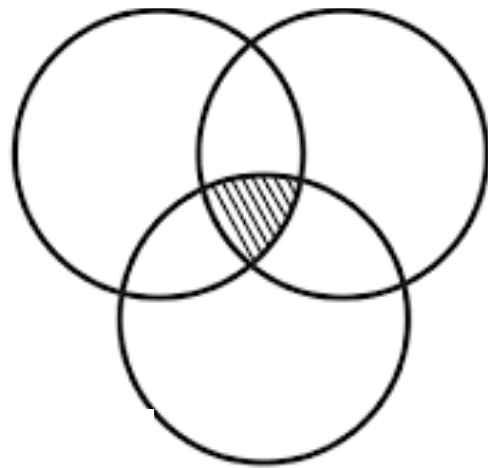
- Give some equivalent statement forms for $(p \wedge q) \vee (s \vee t)$

Associative Law

Associative laws

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$



- Think about an equivalent form for $(p \wedge q) \vee (s \vee t)$

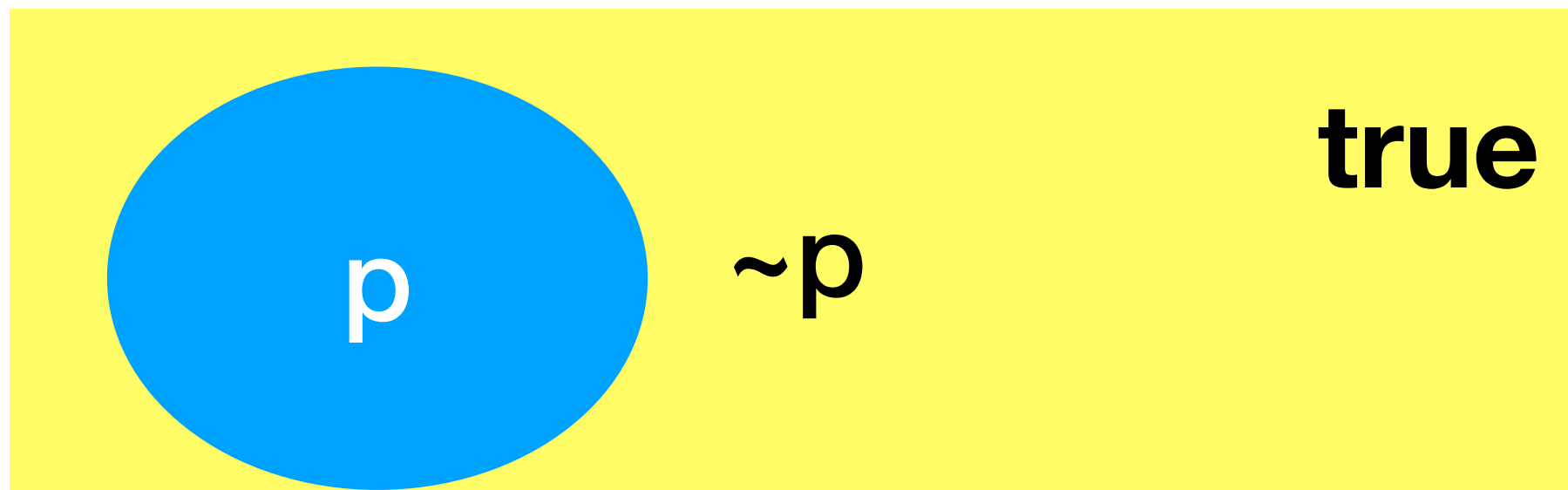
Distributive Law

Distributive laws $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

- A bit like $a * (b + c) = a * b + a * c$
- Think about an equivalent forms for $(p \wedge q) \vee (s \vee t)$

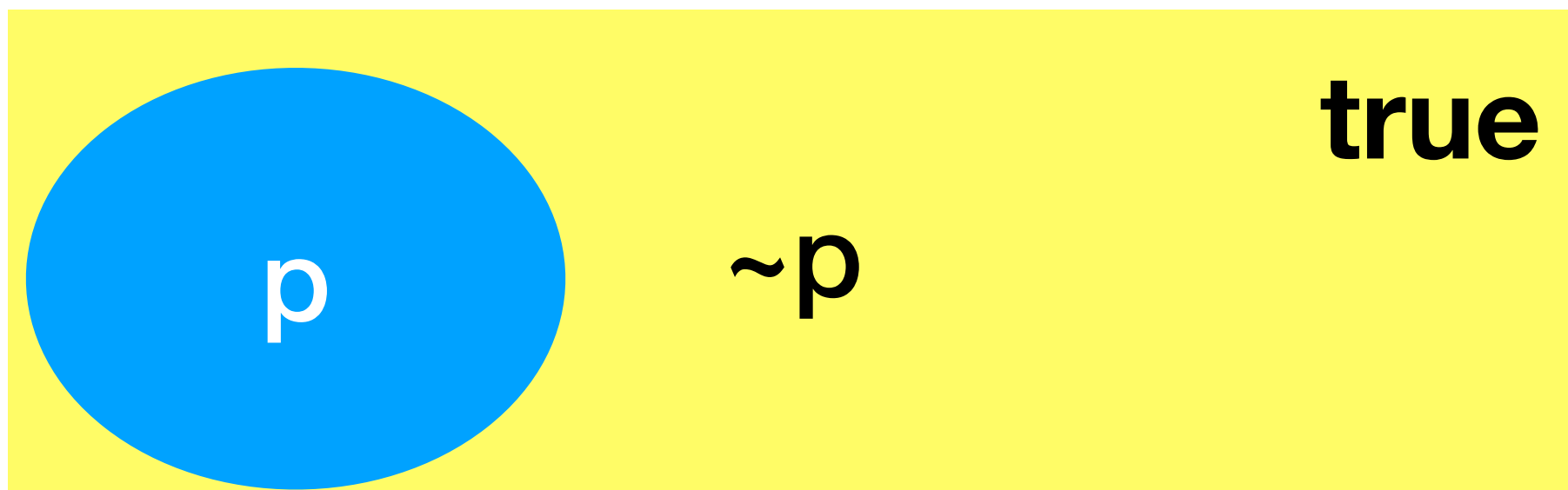
Laws with “true” and “false”

Identity laws	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
Negation laws	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
Uni. bound laws	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$



Double-negation Law

Double neg. law	$\sim (\sim p) \equiv p$
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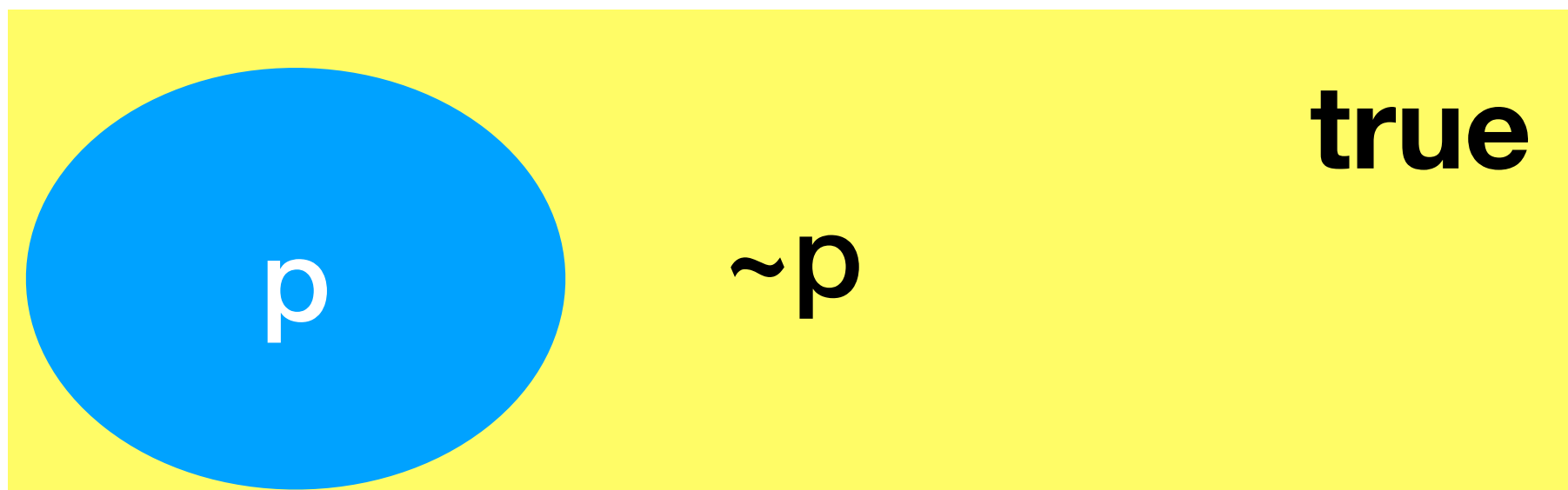


Idempotent Law

Idempotent laws

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$



De Morgan Law

De Morgan's laws	$\sim (p \wedge q) \equiv \sim p \vee \sim q$	$\sim (p \vee q) \equiv \sim p \wedge \sim q$
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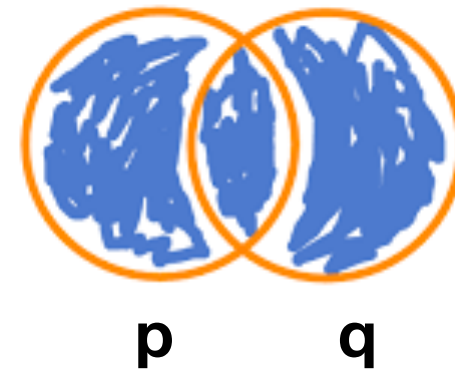
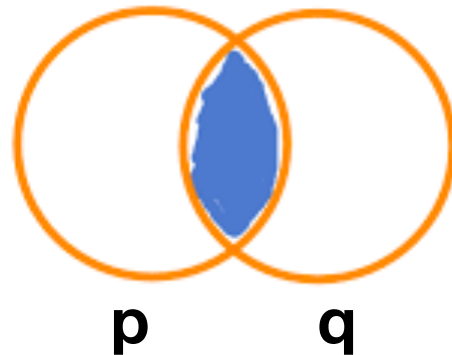
- p = student A is from Korea
- q = student B is from Korea
- $p \wedge q$ = Both student A and B are from Korea
- $\sim(p \wedge q)$ = Either A is not from Korea, or B is not from Korea
- $p \vee q$ = student A or student B is from Korea
- $\sim(p \vee q)$ = _____

Absorption Law

Absorption laws

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$



Equivalence laws

Laws	Formula	Formula
Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity laws	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
Negation laws	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
Double neg. law	$\sim(\sim p) \equiv p$	
Idempotent laws	$p \wedge p \equiv p$	$p \vee p \equiv p$
Uni. bound laws	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
De Morgan's laws	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
Absorption laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negations	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Exercise

Problem 3. [5 points]

Mention whether the following statements are true or false. Reasons are not needed.

- (a) [1 point] $p \vee \sim p \equiv \mathbf{c}$
- (b) [1 point] $p \vee (p \wedge q) \equiv p \wedge (p \vee q)$
- (c) [1 point] $\mathbf{c} \equiv p \vee \mathbf{t}$

Notation: **c** means contradiction, or False
t means True

2021-Final

Cont.

(c) [1 point] $\mathbf{c} \equiv p \vee \mathbf{t}$

(d) [1 point] $p \wedge p \equiv p \vee p$

(e) [1 point] $p \wedge \mathbf{c} \equiv \sim \mathbf{t}$

Exercise

Problem 2. [5 points]

Is conditional operator $\rightarrow$ an associative operator? That is, is $(p \rightarrow q) \rightarrow r$ logically equivalent to $p \rightarrow (q \rightarrow r)$? Prove your answer.

A solution for #2

Problem 2. [5 points]

Is conditional operator $\rightarrow$ an associative operator? That is, is $(p \rightarrow q) \rightarrow r$ logically equivalent to $p \rightarrow (q \rightarrow r)$? Prove your answer.

- $(p \rightarrow q) \rightarrow r = (\sim p \vee q) \rightarrow r = \sim(\sim p \vee q) \vee r = (p \wedge \sim q) \vee r$
- $p \rightarrow (q \rightarrow r) = \sim p \vee (q \rightarrow r) = \sim p \vee (\sim q \vee r) = (\sim p \vee \sim q) \vee r$
- To show the two differ, consider $r = \text{false}$, $\sim q = \text{false}$, $p = \text{false}$
- Alternatively, we could use a truth table

Problem 2. [5 points]

Check the logical equivalence of $((p \wedge q) \rightarrow r)$ and $((p \rightarrow r) \vee (q \rightarrow r))$.

Valid Arguments

<https://youglish.com/pronounce/%22Valid%20Argument%22/english/us?>

Final, 2020-1

Problem 1. [5 points]

Determine if the following deduction rule is valid.

$$p \rightarrow (q \vee r)$$

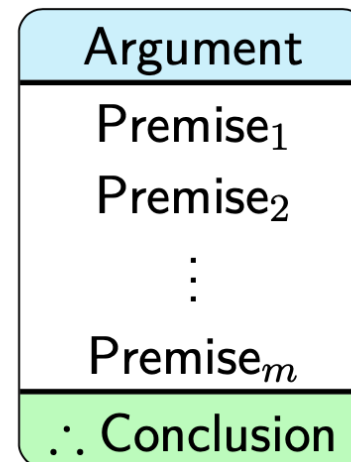
$$\sim (p \rightarrow q)$$

$$\therefore r$$

What is a logical argument?

Definitions

- **Logical argument.** Sequence of statements aimed at demonstrating the truth of an assertion
- **Conclusion.** Last statement in an argument
- **Premises.** Last-but-one statements in an argument



If Premise₁ and Premise₂ and $\dots$ and Premise_m, then Conclusion.

What is a valid argument?

Definition

- An argument is **valid** if the conclusion follows necessarily from the premises

- Every person will die
- Socrates is a person
- So, Socrates will die
- Congressmen/women own classified info
- Investors can take profit from classified info
- So, congressmen/women should not be allowed to actively do investment

Exercise: Valid or Not?

- All cups are blue
- Socrates is cup
- So, Socrates is blue

Valid or not?

Examples

Valid	• If Socrates is a man, then Socrates is mortal. Socrates is a man. Therefore, Socrates is mortal.	$\text{If } p, \text{ then } q.$ $p.$ Therefore, $q.$
Invalid	• If Socrates is a man, then Socrates is mortal. Socrates is mortal. Therefore, Socrates is a man.	$\text{If } p, \text{ then } q.$ $q.$ Therefore, $p.$
Valid	• If Socrates is a man, then Socrates is mortal. Socrates is not mortal. Therefore, Socrates is not a man.	$\text{If } p, \text{ then } q.$ $\sim q.$ Therefore, $\sim p.$
Invalid	• If Socrates is a man, then Socrates is mortal. Socrates is not a man. Therefore, Socrates is not mortal.	$\text{If } p, \text{ then } q.$ $\sim p.$ Therefore, $\sim q.$

How to mathematically check if an argument is valid?

- Truth table
- Inference rules

Method 1: Truth table

1. Identify the premises and conclusion
2. Construct a truth table for premises and conclusion
3. A row of the **truth table** in which all the premises are true is called a **critical row**.

If there is a critical row in which the conclusion is false, then the argument is **invalid**. If the conclusion in every critical row is true, then the argument is **valid**.

Importantly, if there are no critical rows, then the arguments is considered valid

Example

Problem

- Determine the validity of the argument:

$$p \rightarrow q \vee \sim r$$

$$q \rightarrow p \wedge r$$

$$\therefore p \rightarrow r$$

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge r$	$p \rightarrow q \vee \sim r$	$q \rightarrow p \wedge r$	$p \rightarrow r$
T	T	T	F	T	T	T	T	T
T	T	F	T	T	F	T	F	
T	F	T	F	F	T	F	T	
T	F	F	T	T	F	T	T	F
F	T	T	F	T	F	T	F	
F	T	F	T	T	F	T	F	
F	F	T	F	F	F	T	T	T
F	F	F	T	T	F	T	T	T

Exercise

Problem 1. [5 points]

Determine if the following deduction rule is valid.

$$p \rightarrow (q \vee r)$$

$$\sim (p \rightarrow q)$$

$$\therefore r$$

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Inference Rules

Inference rules

Definition

- A **rule of inference** is a valid argument form that can be used to establish logical deductions

Modus Ponens

Definition

- It has the form:
If p , then q
 p
 $\therefore q$
- The term *modus ponens* in Latin means “method of affirming”

p	q	$p \rightarrow q$	p	q
T	T	T	T	T
T	F	F	T	
F	T	T	F	
F	F	T	F	

What is wrong?

- You are a detective investigating a crime scene with your partner, Sherlock. You come across a note that says: "If the butler did it, then the knife will be found in the kitchen."
- Sherlock responds, "Well, we just found the knife in the kitchen. Therefore, the butler did it."

Modus Tollens

Definition

- It has the form:
If p , then q
 $\sim q$
 $\therefore \sim p$
- The term *modus tollens* in Latin means “method of denying”

p	q	$p \rightarrow q$	$\sim q$	$\sim p$
T	T	T	F	
T	F	F	T	
F	T	T	F	
F	F	T	T	T

- If it rains, then the ground will be wet.”
- The ground is not wet
- $\implies ?$

Generalization

Definition

- It has the form:

p

$\therefore p \vee q$

p	q	p	$p \vee q$
T	T	T	T
T	F	T	T
F	T	F	
F	F	F	

Example

- 35 is odd.
 $\therefore$ (more generally) 35 is odd or 35 is even.

What is wrong?

- a friend who tells you, "I really like pizza." Based on this statement, you can apply the Generalization inference rule to make the following conclusion:
- "All people like pizza."

Specialization

Definition

- It has the form:

$$p \wedge q$$

$$\therefore p$$

p	q	$p \wedge q$	p
T	T	T	T
T	F	F	
F	T	F	
F	F	F	

Example

- Ana knows numerical analysis and Ana knows graph algorithms.
 $\therefore$ (in particular) Ana knows graph algorithms

Conjunction

Definition

- It has the form:

p

q

$\therefore p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	
F	T	
F	F	

Example

- Lily loves mathematics.
Lily loves algorithms.
 $\therefore$ Lily loves both mathematics and algorithms.

Elimination

Definition

- It has the form:

$$p \vee q$$

$$\sim q$$

$$\therefore p$$

- Intuition: When you have only two possibilities and you can rule one out, the other must be the case

p	q	$p \vee q$	$\sim q$	p
T	T	T	F	T
T	F	T	T	T
F	T	T	F	F
F	F	F	T	F

Example

- Suppose $x - 3 = 0$ or $x + 2 = 0$.

Also, suppose x is nonnegative.

$$\therefore x = 3.$$

Transitivity

Definition

- It has the form:

$$p \rightarrow q$$

$$q \rightarrow r$$

$$\therefore p \rightarrow r$$

- Can be generalized to a chain with any number of conditionals

Example

- If 18,486 is divisible by 18, then 18,486 is divisible by 9.
If 18,486 is divisible by 9, then the sum of the digits of 18,486 is divisible by 9.
 $\therefore$ If 18,486 is divisible by 18, then the sum of the digits of 18,486 is divisible by 9.

Division into cases

Definition

- It has the form:

$$p \vee q$$

$$p \rightarrow r$$

$$q \rightarrow r$$

$$\therefore r$$

Example

- x is positive or x is negative.
If x is positive, then $x^2 > 0$.
If x is negative, then $x^2 > 0$.
 $\therefore x^2 > 0$.

Summary of Inference rules

Name	Rule	Name	Rule
Modus Ponens	$p \rightarrow q$ p $\therefore q$	Elimination	$p \vee q$ $\sim q$ $\therefore p$
Modus Tollens	$p \rightarrow q$ $\sim q$ $\therefore \sim p$	Transitivity	$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$
Proof by division into cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$	Generalization	p $\therefore p \vee q$
Conjunction	p q $\therefore p \wedge q$	Specialization	q $\therefore p \vee q$ $p \wedge q$ $\therefore p$ $\therefore q$
		Contradiction	$\sim p \rightarrow c$ $\therefore p$

Some wrong inference

Definition

- A **fallacy** is an error in reasoning that results in an invalid argument

Fallacy: Converse error

Definition

- It has the form:

$$p \rightarrow q$$

$$q$$

$$\therefore p$$

- Superficially resembles modus ponens but is invalid

p	q	$p \rightarrow q$	q	p
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	F

Example

- If $x > 2$, then $x^2 > 4$.

$$x^2 > 4.$$

$$\therefore x > 2.$$

Fallacy: Inverse error

Definition

- It has the form:

$$p \rightarrow q$$

$$\sim p$$

$$\therefore \sim q$$

- Superficially resembles modus tollens but is invalid

p	q	$p \rightarrow q$	$\sim p$	$\sim q$
T	T	T	F	F
T	F	F	F	T
F	T	T	T	F
F	F	T	T	T

Example

- If $x > 2$, then $x^2 > 4$.

$$x \leq 2.$$

$$\therefore x^2 \leq 4.$$

Exercises

1. Supply the missing statements using inference rules

1.	1. $p \rightarrow \sim q$ 2. p 3. - - - -	Premise Premise 1,2 Modus Ponens	2.	1. $\sim p \rightarrow q$ 2. $\sim p$ 3. - - - -	Premise Premise 1,2 Modus Ponens
3.	1. $(\sim p \vee q) \rightarrow \sim(q \wedge r)$ 2. $\sim p \vee q$ 3. - - - -	Premise Premise 1,2 Modus Ponens	4.	1. $(\sim p \wedge q) \rightarrow (q \wedge \sim r)$ 2. $\sim p \wedge q$ 3. - - - -	Premise Premise 1,2 Modus Ponens
5.	1. $(\sim p \vee q) \rightarrow \sim(q \wedge r)$ 2. $q \wedge r$ 3. - - - -	Premise Premise 1,2 Modus Tollens	6.	1. $(\sim p \wedge q) \rightarrow (q \wedge \sim r)$ 2. $\sim(q \wedge \sim r)$ 3. - - - -	Premise Premise 1,2 Modus Tollens
7.	1. $\sim(\sim p \vee q)$ 2. - - - -	Premise De Morgan	8.	$\sim(p \wedge \sim q)$ 2. - - - -	Premise De Morgan
9.	1. $(p \wedge r) \rightarrow \sim q$ 2. $\sim q \rightarrow r$ 3. - - - -	Premise Premise 1,2 Transitive Law	10.	1. $(\sim p \wedge q) \rightarrow (q \wedge \sim r)$ 2. $(q \wedge \sim r) \rightarrow s$ 3. - - - -	Premise Premise 1,2 Transitive Law

2.	1. $(p \wedge r) \rightarrow \sim q$	Premise	12.	1. $(\sim p \wedge q) \rightarrow (q \wedge \sim r)$	Premise
	2. $\sim q \rightarrow r$	Premise		2. $(q \wedge \sim r) \rightarrow s$	Premise
	3. $\sim r$	Premise		3. $\sim s$	Premise
	4. - - - -	1,2 Transitive Law		4. - - - -	1,2 Transitive Law
	5. - - - -	3,4 Modus Tollens		5. - - - -	3,4 Modus Tollens

15.	1. $p \rightarrow (r \wedge q)$	Premise
	2. $\sim r$	Premise
	3. - - - -	2, Addition of $\sim q$
	4. - - - -	3, De Morgan
	5. - - - -	1,4 Modus Tollens

17.	1. $(p \wedge q) \rightarrow r$	Premise	18.	1. $p \rightarrow r$	Premise
	2. q	Premise		2. p	Premise
	3. p	Premise		3. s	Premise
	4. - - - -	3,2 Rule C		4. - - - -	1,2 Modus Ponens
	5. - - - -	1,4 Modus Ponens		5. - - - -	3,4 Rule C

These great exercises are taken from
<https://www.zweigmedia.com/RealWorld/logic/logicex5.html>

Prove the following is valid
using logical inference

$$\begin{array}{l} p \rightarrow (q \vee r) \\ \sim (p \rightarrow q) \\ \therefore r \end{array}$$

Prove the following is valid
using logical inference

a. $\sim p \vee q \rightarrow r$

b. $s \vee \sim q$

c. $\sim t$

d. $p \rightarrow t$

e. $\sim p \wedge r \rightarrow \sim s$

f. $\therefore \sim q$

Prove the following is valid
using logical inference

a. $p \vee q$

b. $q \rightarrow r$

c. $p \wedge s \rightarrow t$

d. $\sim r$

e. $\sim q \rightarrow u \wedge s$

f. $\therefore t$

Problem of truth tellers and liars

Problem

- There is an island containing two types of people: truth tellers who always tell the truth and liars who always lie. You visit the island and are approached by two natives who speak to you as follows:
 A says: *B* is a truth teller.
 B says: *A* and I are of opposite type.
- What are *A* and *B*?

**Education is what remains after one has forgotten
what one has learned in school. — Albert Einstein**